

Time Series Forecasting of Energy Commodity Prices Using Macroeconomic Indicators and GDELT Derived News Signals

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Abstract

Accurate forecasting of energy commodity prices — particularly West Texas Intermediate (WTI) crude oil and Henry Hub natural gas — remains a persistent methodological challenge, owing to their intrinsic price volatility, pronounced sensitivity to geopolitical disruptions, and structurally complex dependence on macroeconomic cycles. This paper examines whether macroeconomic indicators from the Federal Reserve Economic Data (FRED) database and news-based signals derived from the Global Database of Events, Language, and Tone (GDELT) improve forecasting performance relative to standard price-only baselines. The analysis covers a full forecasting pipeline, including stationarity testing, seasonal decomposition, feature transformation, and multivariate modeling with both classical statistical methods and neural architectures with exogenous inputs. We tuned hyperparameters with Optuna using 30 trials for each model configuration. We evaluated model performance with Root Mean Squared Error (RMSE) and further checked it with walk-forward backtesting. Empirical results indicate that, while point-level forecast accuracy remained constrained across all configurations, the proposed multivariate framework exhibited a meaningful capacity to identify directional price movements and cyclical turning points that univariate baseline models systematically failed to detect. Collectively, these findings suggest that heterogeneous data fusion holds substantive promise for US energy price modeling, while simultaneously underscoring the practical limitations of aggregated news-derived features in high-frequency commodity forecasting contexts.

Keywords: Time Series, GDELT, News Sentiment, Macroeconomic Indicators, FRED, Energy Price Forecasting, Oil, Natural Gas

1 Introduction

Energy commodity prices — most prominently crude oil and natural gas — serve as first-order indicators of global economic activity and are deeply embedded in the transmission mechanisms of inflation, industrial output, and monetary policy [4, 15, 19]. Changes in these prices affect transportation costs, household energy bills, production decisions, and inflation expectations, so even modest improvements in forecasting can matter for policymakers, firms, and investors [4, 15]. At the same time, forecasting is difficult because price movements reflect a shifting mix of supply imbalances, geopolitical disruptions, business-cycle conditions, and financial-market behavior [14, 19].

Energy-price forecasting has therefore been studied through both classical time-series models and more flexible neural architectures, but no single model family consistently dominates across commodities and market regimes [5, 16, 18, 26]. In parallel, prior work suggests that macroeconomic context and news-driven information may add signal beyond price history alone, motivating comparative evaluation of structured external features alongside standard univariate baselines [19, 24, 27, 28].

Still, work that evaluates price history, macroeconomic variables, and news-based signals within one shared forecasting framework is relatively limited [4, 27]. Univariate models capture persistence in the series but ignore outside information. Multivariate models are more informative, but their gains are not always stable or easy to compare across studies [4, 5, 18]. Many papers also focus on one commodity, one model family, or one external data source at a time. That makes it hard to judge the relative value of macroeconomic indicators and event-driven news under the same evaluation design. To address that issue, we build a reproducible forecasting framework that compares classical statistical models and hybrid neural architectures across four settings: univariate, macro-only, news-only, and combined features. We evaluate all models with the same walk-forward validation logic and the same error metrics.

The main expectation is that external information should be most useful when macroeconomic indicators and news-derived signals are considered together rather than separately. More specifically, we test whether macroeconomic variables improve upon univariate baselines, whether GDELT-derived signals contribute additional information, and whether the joint feature space produces stronger directional performance than either feature group on its own.

This paper makes three contributions. First, it compares univariate and multivariate forecasting strategies for both WTI crude oil and Henry Hub natural gas within a single pipeline. Second, it separates the contribution of macroeconomic variables, news-derived variables, and their combination so that the role of each feature group is easier to interpret. Third, it pairs standard forecast metrics with walk-forward backtesting to check whether apparent improvements persist beyond a single held-out period.

2 Related Work

The literature on energy-price forecasting can be organized into three broad strands: classical econometric work on price dynamics and structural shocks, deep-learning studies that emphasize nonlinear prediction, and news- or text-based work that brings in external information. Because this paper compares macroeconomic and GDELT-derived inputs inside one common pipeline, the most relevant prior studies are not just those with the lowest reported errors, but those that clarify what each source of information adds, under what evaluation design, and where the evidence is still thin.

Classical energy-price and macroeconomic forecasting studies

Early empirical work made it clear that oil-price movements cannot be treated as a purely mechanical autoregressive process. Hamilton [14] showed that oil shocks emerge through distinct economic channels and are tied to broader macroeconomic adjustment. Kilian [19] pushed this further by separating oil-price fluctuations into demand- and supply-driven shocks, showing that price changes do not all have the same origin or implication. These studies matter here because they justify using exogenous information instead of relying only on lagged prices. At the same time, they were not designed as broad forecasting comparisons across multiple model classes, nor do they ask whether modern event-based data improve practical out-of-sample prediction for both crude oil and natural gas.

Baumeister and Kilian [4] offered a useful cautionary perspective. Their review of four decades of oil-price behavior argues that oil prices remain hard to predict because structural relationships shift over time and because apparent forecasting gains often fade across regimes. That point directly motivates the walk-forward backtesting used in this paper. Even so, their article is a macroeconomic synthesis rather than a side-by-side empirical comparison of classical, neural, macro-only, news-only, and combined-feature models within one reproducible framework.

Within the time-series tradition, Box and Jenkins [5] established the foundations of ARIMA-type forecasting, while Durbin and Koopman [10] formalized state-space methods for modeling latent dynamics and measurement structure. These references remain central because they justify ARIMA-, SARIMA-, and SARIMAX-type baselines before moving to more flexible models. What they do not answer is the specific question considered here: once those baselines are in place, do macroeconomic indicators and event-derived news features actually improve forecasting?

A closer benchmark for this project is the use of macroeconomic covariates in energy-price prediction. FRED itself is data infrastructure rather than a forecasting study [11], but the variables it distributes - industrial production, exchange rates, inflation proxies, and volatility measures - are exactly the kinds of structured signals commonly used to contextualize commodity prices. The limitation of this macro-focused literature is that it usually stops there. It seldom asks how macroeconomic variables perform alongside event-driven external signals, and it rarely studies both WTI crude oil and Henry Hub natural gas under the same evaluation design.

Deep-learning studies for commodity and financial forecasting

A second line of work asks whether neural architectures can outperform classical statistical models when the underlying process is nonlinear. Hochreiter and Schmidhuber [16], together with Gers et al. [12], developed the LSTM architecture to handle long-range temporal dependence more effectively than standard recurrent networks. Those papers justify the use of LSTM here, but they are architecture papers rather than direct commodity-forecasting benchmarks.

Application-oriented studies report a more mixed picture. Cen and Wang [7] used an LSTM-based framework for crude-oil prediction and argued that deep learning can help when prior knowledge and transferred representations are incorporated. Busari and Lim [6] compared AdaBoost-LSTM and AdaBoost-GRU models for crude oil and found gains over simpler baselines. Kim and Won [20], working on stock-index volatility rather than energy markets, likewise showed that hybrid deep-learning pipelines can outperform narrower benchmarks when complementary temporal representations are combined. Taken together, these studies support the idea that neural models can exploit nonlinear structure.

At the same time, those studies do not fully address the question pursued here. Most focus on one commodity or one financial series rather than comparing two energy commodities with different market structures. They also tend to emphasize point-forecast accuracy without pairing it with walk-forward backtesting. Finally, they usually do not separate the contribution of macroeconomic variables from that of event-driven news within the same experimental design. This paper therefore builds on the deep-learning literature, but asks a broader comparative question: not just whether LSTM-type models can work, but when they help relative to classical models and which kinds of exogenous information matter most.

Hybrid CNN-LSTM architectures are especially relevant in this context. LeCun et al. [23] established the broader logic of convolutional feature extraction, and Sainath et al. [26] showed how convolutional and recurrent components can be combined to exploit both local and sequential structure. That architecture motivates the CNN-LSTM models used here, especially in multivariate settings where short-run feature interactions may matter. Still, prior hybrid-architecture studies leave open the question most important for this paper: whether these models benefit more from macroeconomic context, from news-derived signals, or from the joint feature space.

News, sentiment, and event-driven external signals

A third and increasingly important literature examines whether text or event information improves market prediction. Tetlock [28] showed that media content contains economically relevant information and that textual pessimism is associated with market behavior. Li et al. [25] then showed that news sentiment can improve stock-return prediction, suggesting that narrative information carries incremental value beyond prices alone. More recently, Shapiro et al. [27] developed a systematic approach to measuring news sentiment and showed that such indices can recover interpretable macro-financial information. These studies matter directly here because they support

the idea that event-driven information may help forecast energy prices, or at least improve directional interpretation.

However, much of this literature focuses on equity markets, general macro-financial conditions, or custom text corpora rather than monthly forecasting of WTI crude oil and Henry Hub natural gas. Many text-based approaches also depend on full-text processing or hand-built sentiment dictionaries, which can make replication harder. Our setup differs by using GDELT, a structured event database that provides standardized information on event type, tone, and geopolitical context at scale.

Leetaru and Schrodtt [24] introduced GDELT as a global event database that converts large news flows into structured records. Goldstein [13] provided the conflict-cooperation scale that underlies one of the key event-intensity measures used here. Baker et al. [3] also showed the broader predictive usefulness of text-derived uncertainty measures, reinforcing the view that narrative and event information can encode economically relevant variation. What these studies do not show is whether GDELT-derived features outperform or complement macroeconomic variables in energy-price forecasting under a controlled comparison with ARIMA, SARIMA, SARIMAX, LSTM, and CNN-LSTM models.

Taken together, the literature points to three main conclusions that motivate this project. First, classical studies show that macroeconomic structure matters, but they also warn that forecast gains may not survive across regimes [4, 14, 19]. Second, deep-learning studies suggest that nonlinear models can help, though their gains are often dataset-specific and not always benchmarked against strong classical baselines [6, 7, 20]. Third, news- and event-based studies indicate that external narrative information is useful, but they rarely test its incremental value jointly with macroeconomic variables in a reproducible energy-forecasting setting [24, 25, 27, 28].

Despite these advances, a unified benchmarking framework that simultaneously compares the isolated and joint contributions of macroeconomic and news-based signals — using both test-set accuracy and temporal walk-forward backtesting — remains absent from the literature. The present study addresses this gap by providing such a framework across two distinct energy commodities.

3 Data

This study uses two main data sources: macroeconomic indicators and global news event data. The objective is to integrate structured macroeconomic signals with event-driven geopolitical information in a temporally aligned and methodologically consistent framework, thereby enabling a rigorous assessment of their respective and joint contributions to energy price predictability. Macroeconomic variables capture the broader economic setting, while news-based signals provide a partial view of real-world events, sentiment, and market stress. Using both sources together allows us to examine whether they contribute distinct and complementary information for energy-price prediction.

3.1 FRED Data

The macroeconomic dataset is obtained from the Federal Reserve Economic Data (FRED) platform, which provides a centralized collection of economic time series compiled from official sources. We access the series through the FRED API so that variables with different frequencies can be collected and processed in a consistent way.

The selected macroeconomic variables are meant to represent broad channels through which energy prices may be affected, including industrial activity, inflation expectations, exchange-rate conditions, and financial-market uncertainty. These variables do not mechanically determine oil and gas prices, but they do summarize the economic environment in which demand conditions, investment decisions, and risk perceptions evolve.

3.1.1 Target Variables

The forecasting targets are the two energy price series listed in Table 1.

Table 1: Target variables used in forecasting

Variable Name	FRED Code	Description
Crude Oil Price	DCOILWTICO	Spot price of crude oil
Natural Gas Price	DHHNGSP	Natural gas spot price

The retained macroeconomic block used in the later multivariate models consists of *ind_prod_drill*, *real_usd_index*, *brent_crude_oil*, *cap_util*, and *volatility_index*. Table 2 reports their post-screening VIF values, while Figure 1 provides a compact visual summary of how these retained FRED variables co-move with WTI crude oil and Henry Hub natural gas.

The heatmap helps motivate the macroeconomic feature block in two ways. First, it shows that several retained predictors — especially drilling activity, the real dollar index, and Brent prices — have visible linear association with the target series, which supports their inclusion as economically meaningful covariates rather than arbitrary additions. Second, the predictor block does not appear dominated by near-perfect pairwise dependence, which is consistent with the moderate VIF values in Table 2. In this sense, the heatmap complements the VIF table: the table screens out problematic redundancy, while the figure shows why the remaining FRED variables are still informative enough to justify multivariate modeling.

For the neural-network models, all continuous inputs are normalized before sequence construction so that variables on very different scales do not dominate optimization. This improves training stability and makes the learned representations more comparable across feature groups. Normalization parameters (mean and standard deviation) were computed exclusively on the training partition and subsequently applied to the validation and test sets to prevent data leakage.

Table 2: VIF values
for retained
macroeconomic
features.

Feature	VIF
<i>ind_prod_drill</i>	3.67
<i>real_usd_index</i>	2.47
<i>brent_crude_oil</i>	2.16
<i>cap_util</i>	2.13
<i>volatility_index</i>	1.70

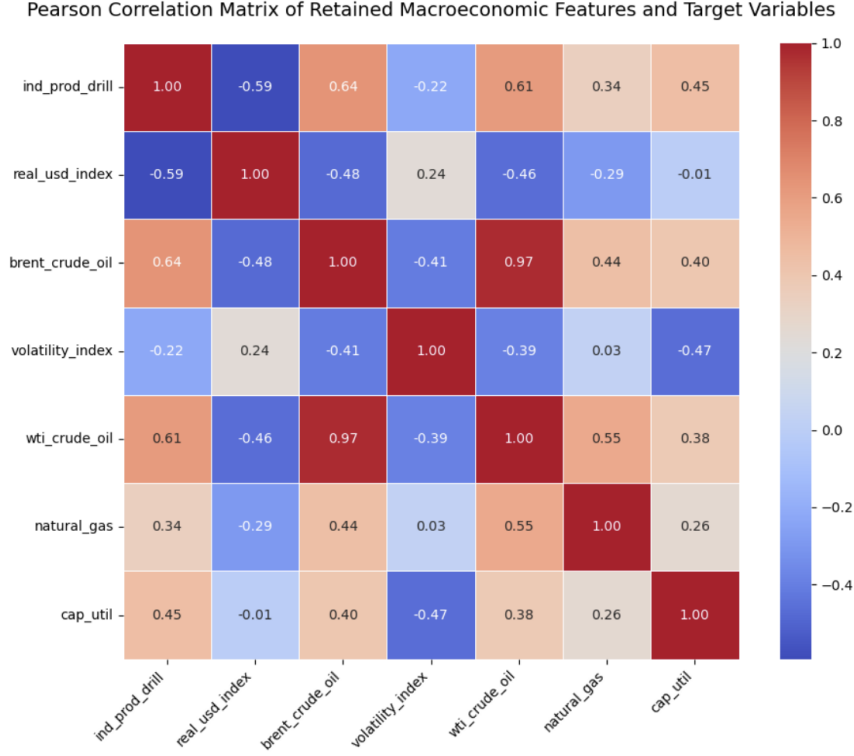


Fig. 1: Pearson correlation heatmap for retained FRED variables and target prices.

3.1.2 GDELT Data Processing

We process the GDELT data to extract a compact set of signals from a very large event archive. The main steps include retaining only US-related events, one-hot encoding ‘EventCode’, aggregating event counts daily, and computing summary measures such as average Goldstein Scale and average tone. When needed, these series are further aggregated to the monthly level. The result is a more compact and interpretable time-series representation of event activity. Procedure 1 formalizes this transformation from raw event records to aggregated forecasting inputs.

Procedure 1 GDELT Data Processing and Feature Extraction

Input: Raw GDELT event records R , target country $C = \text{US}$, target frequency $\text{freq} = \text{monthly}$

Output: Aggregated feature matrix G_{agg} indexed by month

```
1:  $R_{\text{US}} \leftarrow \{r \in R : r.\text{Actor1CountryCode} = C \text{ or } r.\text{Actor2CountryCode} = C\}$ 
2:  $\text{EVENTCATEGORIES} \leftarrow \text{UNIQUEVALUES}(R_{\text{US}}.\text{EventCode})$ 
3: for all category  $c$  in  $\text{EVENTCATEGORIES}$  do
4:    $R_{\text{US}}[\text{EventCode}_c] \leftarrow \mathcal{K}[R_{\text{US}}.\text{EventCode} = c]$ 
5: end for
6:  $G_{\text{agg}} \leftarrow \text{GROUPBYMONTH}(R_{\text{US}})$ 
    $\text{EVENTCOUNT} \leftarrow \sum \text{NumMentions}$ 
    $\text{AVGGOLDSTEIN} \leftarrow \text{mean}(\text{GoldsteinScale})$ 
    $\text{AVGTONE}_{\text{monthly}} \leftarrow \text{mean}(\text{AvgTone})$ 
    $\text{EVENTCODECOUNTS} \leftarrow \sum (\text{EventCode}_* \text{ columns})$ 
7: for all category  $c$  in  $\text{EVENTCATEGORIES}$  do
8:    $G_{\text{agg}}[\text{EventCode}_c] \leftarrow G_{\text{agg}}[\text{EventCode}_c] / G_{\text{agg}}[\text{EventCount}]$ 
9: end for
10: return  $G_{\text{agg}}$ 
```

3.1.3 Data Integration

We then merge the processed macroeconomic and GDELT datasets on a common date index. We retain only the overlapping sample so that all model inputs remain temporally aligned.

At this point, the paper moves from data construction to model-design choices. The next section explains how the aligned dataset is partitioned, transformed, and evaluated so that the later empirical comparisons remain consistent across classical and neural forecasting approaches.

4 Methodology

This section describes the forecasting workflow applied after the FRED and GDELT sources are integrated. We first define the temporal splitting strategy, then outline the preprocessing and stationarity checks used to prepare the inputs, and finally present the classical and neural models together with the evaluation framework. Organizing the pipeline in this order keeps the later results directly connected to the design choices that produced them.

4.1 Data Splitting Strategy

We use different splitting strategies for classical and neural models so that each class is evaluated fairly. For the classical time-series models, the data are divided into training and testing periods. This setup evaluates each model’s ability to forecast future values using only past observations.

The neural models use a three-part split consisting of backtest, train, and test segments. Backtesting is useful here because it helps distinguish genuine temporal

Table 3: Classical model data splitting

Stage	Sequence Range (monthly)	Role
Train	January 2014–October 2025	Model estimation
Test	November 2025–March 2026	Final held-out evaluation

learning from favorable performance on a single slice of data. To keep the rule consistent across configurations, the split is defined using the input-window size alone. For an input window of size k , the backtest segment spans indices 0 through $k + 5$, the held-out test segment spans indices $n - (k + 5)$ through n , and the observations in between are used for training. Table 4 summarizes this ordering.

Table 4: Data splitting and sliding window strategy

Stage	Calendar Order	Sequence Definition	Role
Backtest	First	$0 \rightarrow k + 5$	Rolling validation
Train	Second	$(k + 6) \rightarrow n - (k + 6)$	Model fitting
Test	Final	$n - (k + 5) \rightarrow n$	Held-out evaluation

After integrating the FRED and GDELT sources, we retain a monthly sample that runs from January 2014 through March 2026. For the neural models, the split depends only on input-window size: the first $k + 5$ months form the backtest block, the last $k + 5$ months form the test block, and the middle portion is used for training.

Table 5: Exact calendar ranges used for neural-network train, test, and backtest splits

Input Window Size	Backtest (monthly)	Train (monthly)	Test (monthly)
5	Jan 2014–Oct 2014	Nov 2014–May 2025	Jun 2025–Mar 2026
10	Jan 2014–Mar 2015	Apr 2015–Dec 2024	Jan 2025–Mar 2026
24	Jan 2014–May 2016	Jun 2016–Oct 2023	Nov 2023–Mar 2026
36	Jan 2014–May 2017	Jun 2017–Oct 2022	Nov 2022–Mar 2026

4.2 Key Design Principles

The pipeline follows a few simple design principles. Strict adherence to temporal ordering is enforced at every stage of the pipeline to preserve the causal structure of the time series and to prevent any form of look-ahead bias that would artificially inflate model performance. In accordance with the no-future-information constraint fundamental to out-of-sample forecasting evaluation, all models are trained exclusively on observations preceding the forecast horizon, with strict enforcement applied across both the classical and neural network model classes. Data leakage is avoided throughout, and the preprocessing logic is kept as consistent as possible across model families so that the comparisons remain fair.

Overall, this preprocessing pipeline provides a consistent basis for integrating macroeconomic indicators and news-based signals while keeping the later model comparisons interpretable.

4.3 Stationarity Testing

We test for stationarity before fitting the classical time-series models, since ARIMA-type specifications assume that the mean and variance structure of the modeled series is sufficiently stable over time. To do this, we use both the Augmented Dickey–Fuller (ADF) test and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test. The ADF test takes the null hypothesis that the series contains a unit root and is therefore non-stationary, while the KPSS test reverses the perspective and treats stationarity as the null. Using both tests provides a more balanced diagnostic before model fitting. Throughout the analysis, we use a significance level of $\alpha = 0.05$. The ADF regression is given by

$$\Delta y_t = \alpha + \beta t + \phi y_{t-1} + \sum_{i=1}^p \gamma_i \Delta y_{t-i} + \epsilon_t, \quad (1)$$

The KPSS test statistic is computed as

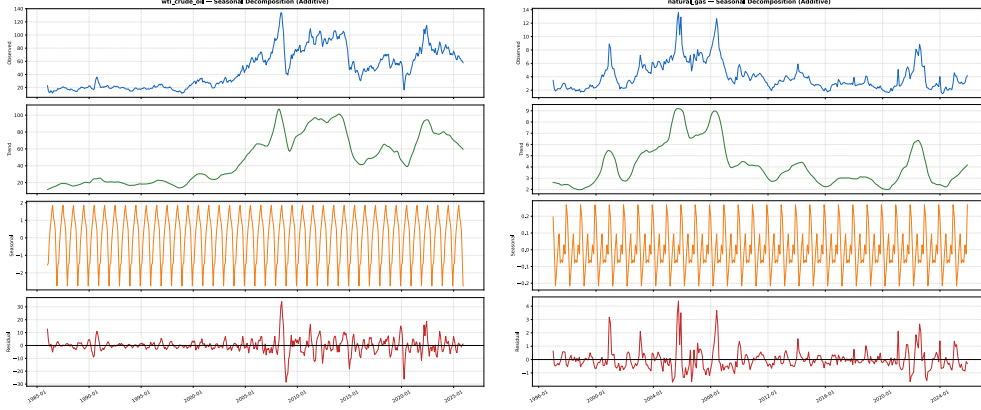
$$\text{KPSS} = \frac{1}{T^2 \hat{\sigma}^2} \sum_{t=1}^T S_t^2, \quad (2)$$

where $S_t = \sum_{i=1}^t e_i$ denotes the cumulative partial sum of the residuals from the regression used in the KPSS procedure, T is the sample size, and $\hat{\sigma}^2$ is the long-run variance estimate. In Equation 1, α is a constant, βt is a deterministic trend term, ϕ is the coefficient on the lagged level term y_{t-1} , p is the lag order, and ϵ_t is a white-noise error. When the ADF null cannot be rejected or the KPSS null is rejected, the series is differenced and tested again until stationarity is obtained. The required number of differences then determines the integration order d in the corresponding ARIMA(p, d, q) specification. In practice, the level series generally did not satisfy the combined stationarity criteria, but the transformed series used for classical modeling satisfied both tests at the 5% level after differencing.

To complement the formal stationarity diagnostics, we also inspect seasonal-trend decomposition using Loess (STL) for both target series. STL separates each monthly price series into trend, seasonal, and residual components, which provides a visual diagnostic of recurring within-year structure alongside slower-moving medium-run dynamics. As shown in Figure 2, both WTI crude oil and Henry Hub natural gas exhibit nontrivial seasonal structure, which helps justify the inclusion of seasonal differencing and seasonal lag terms in the SARIMA and SARIMAX specifications.

4.4 Modeling

This study uses both classical statistical models and neural network-based approaches so that different aspects of the data can be evaluated under the same framework. Classical models serve as transparent baselines, while neural models are included to capture possible nonlinear relationships, interactions among variables, and longer temporal dependencies.



(a) WTI crude oil decomposition

(b) Henry Hub natural gas decomposition

Fig. 2: STL decomposition of the monthly target price series into observed, trend, seasonal, and residual components. The decompositions provide visual evidence of seasonality and medium-run structure that motivate the seasonal terms used in the SARIMA and SARIMAX models.

4.5.1 Classical Modeling

4.5.1.1 ARIMA

In ARIMA, the model first transforms the time series into a stationary form through differencing. Then, for each time step t , the model computes the predicted value by combining a weighted sum of previous observations and previous error terms, as defined by,

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t. \quad (3)$$

Here, the algorithm iteratively learns coefficients ϕ_i and θ_j , which determine how much influence past values and past errors have on the current prediction. This makes ARIMA effective for modeling linear temporal dependencies in univariate data.

4.5.1.2 SARIMA

SARIMA follows the same process as ARIMA but extends it by incorporating seasonal patterns. After differencing both regular and seasonal components, the model applies seasonal autoregressive and moving average terms:

$$\Phi_P(B^s)\phi_p(B)(1-B)^d(1-B^s)^D y_t = \Theta_Q(B^s)\theta_q(B)\epsilon_t. \quad (4)$$

In this formulation, the algorithm represents repeating cycles by applying operations at lag intervals defined by the seasonal period s . This allows the model to account for periodic fluctuations observed in energy price data.

4.5.1.3 SARIMAX

SARIMAX extends SARIMA by integrating external variables into the prediction step. At each time step, the model not only considers past values and errors but also includes the contribution from exogenous variables:

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \beta X_t + \epsilon_t. \quad (5)$$

Here, the algorithm learns an additional set of coefficients β that quantify the influence of macroeconomic and news-based features X_t . This enables the model to incorporate real-world drivers of price movement, making it more suitable for multivariate forecasting.

4.5.2 Neural Network-Based Modeling

Neural network models learn patterns directly from data by optimizing weights through iterative training. Instead of explicitly defining relationships, they approximate complex mappings between input sequences and outputs.

4.5.2.1 LSTM

In LSTM, the model processes the input sequence step by step, where each input x_t and previous hidden state h_{t-1} are used to update the memory cell. At each step, the algorithm computes gate values to control information flow:

$$\begin{aligned} f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f), \\ i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i), \\ o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o), \\ \tilde{C}_t &= \tanh(W_c[h_{t-1}, x_t] + b_c). \end{aligned} \quad (6)$$

The cell state is then updated as

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t, \quad (7)$$

and the hidden state is computed as

$$h_t = o_t \cdot \tanh(C_t). \quad (8)$$

This step-by-step process allows the model to selectively retain or forget information across time steps. Finally, the last hidden state is passed through a linear layer,

$$\hat{y} = Wh_t + b, \quad (9)$$

to generate the forecast output. This mechanism enables LSTM to model long-term dependencies in sequential data. In implementation, the LSTM receives input tensors

of shape (batch, time, features) and maps the final hidden representation to a continuous forecast through a linear output layer, making it suitable for nonlinear multivariate sequence learning.

4.5.2.2 CNN–LSTM

Fig. 3: CNN–LSTM pipeline.

The CNN–LSTM model follows a two-stage process. First, the input sequence is passed through a convolutional layer,

$$z = \text{Conv1D}(x), \quad (10)$$

which extracts local temporal features by applying filters across time steps. This transforms the input into feature maps that highlight short-term patterns. Next, these extracted features are passed into the LSTM layer,

$$h_t = \text{LSTM}(z_t), \quad (11)$$

where the LSTM processes the sequence of feature maps to model long-term dependencies. Finally, the output is computed using

$$\hat{y} = Wh_t + b. \quad (12)$$

Figure 3 shows the CNN–LSTM architecture used in this study. This combination allows the model to first identify local structures and then learn their evolution over time, making it particularly effective for complex time series influenced by multiple factors. As with the LSTM, the CNN–LSTM operates on tensors of shape (batch, time, features), with the final recurrent representation passed to a linear layer for continuous-value prediction.

The model-training and hyperparameter-selection routine for this architecture is formalized in Procedure 2.

Procedure 2 CNN-LSTM Training and Hyperparameter Selection

Input: Training sequences $(X_{\text{seq}}, Y_{\text{seq}})$ of shape (N, k, F) ; number of Optuna trials $T = 30$; validation split $\rho = 0.1$

Output: Selected CNN-LSTM model m^* and hyperparameter vector θ^*

```
1: Define search space  $\Theta$ : convolution filters  $\in \{32, 64\}$ , kernel size  $\in \{2, 3, 5\}$ ,  
   LSTM units  $\in \{32, 64\}$ , dropout rate  $\in [0.1, 0.3]$ ,  $L_2$  regularization  $\in [10^{-4}, 10^{-2}]$ ,  
   learning rate  $\in [10^{-4}, 3 \times 10^{-3}]$ , and batch size = 32  
2:  $\ell^* \leftarrow \infty$   
3: for  $t = 1$  to  $T$  do  
4:    $\theta_t \leftarrow \text{OPTUNASUGGEST}(\Theta)$   
5:   Construct candidate model  $m_t$  with a Conv1D layer, an LSTM layer, dropout  
   regularization, and a dense output layer parameterized by  $\theta_t$   
6:   Compile  $m_t$  with the Adam optimizer and Huber loss  
7:   Train  $m_t$  on  $(X_{\text{seq}}, Y_{\text{seq}})$  using validation split  $\rho$  and early stopping based on  
   validation loss  
8:    $\ell_t \leftarrow \min(\text{VALIDATIONLOSSHISTORY}(m_t))$   
9:   if  $\ell_t < \ell^*$  then  
10:      $\ell^* \leftarrow \ell_t$   
11:      $\theta^* \leftarrow \theta_t$   
12:      $m^* \leftarrow m_t$   
13:   end if  
14: end for  
15: return  $m^*, \theta^*$ 
```

4.5.3 Hyperparameter Tuning

To improve model performance, we optimize hyperparameters using Optuna. For each neural-network configuration, the study is run for 30 trials with optimization direction set to minimization. In the CNN-LSTM implementation, the Optuna objective returns the minimum validation loss achieved during training, i.e., `min(val_loss)` across epochs, so model selection is based on the best validation-set performance rather than on the training loss. The optimizer used during model fitting is Adam with Huber loss, while early stopping monitors validation loss to reduce overfitting and restore the best-performing weights.

The search space used in the CNN-LSTM tuning routine is summarized in Table 6. Because the Optuna study is instantiated with `optuna.create_study(direction="minimize")` and no custom sampler or pruner is supplied, the implementation uses Optuna’s default Tree-structured Parzen Estimator (TPE) sampler and does not apply pruning. Batch size is fixed at 32 during tuning, and sequence construction follows the same window specification used for the corresponding training configuration rather than being tuned inside the Optuna objective.

This search process ensures that the neural models are trained under systematically varied settings across the univariate, macro-only, news-only, and combined-input feature sets, while keeping the tuning protocol consistent across experiments. The

Table 6: Hyperparameter search space used for CNN-LSTM tuning.

Hyperparameter	Range/Options	Type
Convolution filters	{32, 64}	Categorical
Kernel size	{2, 3, 5}	Categorical
LSTM units	{32, 64}	Categorical
Dropout rate	[0.1, 0.3]	Continuous
L_2 regularization	$[10^{-4}, 10^{-2}]$	Log-continuous
Learning rate	$[10^{-4}, 3 \times 10^{-3}]$	Log-continuous
Batch size	32	Fixed

exact CNN-LSTM tuning workflow referred to throughout this subsection is given in Procedure 2.

4.5.4 Evaluation and Backtesting

We evaluate model performance using three standard error measures: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). RMSE penalizes large errors more strongly and is useful when large forecast deviations are especially undesirable. MAE provides an interpretable average magnitude of the forecast error. MAPE expresses the error in percentage terms, which helps compare model performance across crude oil and natural gas despite their different price scales.

The metrics are defined as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}, \quad (13)$$

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|, \quad (14)$$

$$\text{MAPE} = \frac{100}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right|. \quad (15)$$

To assess whether observed differences in forecast accuracy between model pairs are statistically significant rather than attributable to sampling variability, the Diebold-Mariano (DM) test [9] is applied. Let the loss differential be defined as

$$d_t = L(e_{1t}) - L(e_{2t}), \quad (16)$$

where $L(\cdot)$ is a loss function and, in this study, squared error is used. The DM statistic is then given by

$$\text{DM} = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})/n}} \sim \mathcal{N}(0, 1), \quad (17)$$

under the null hypothesis of equal predictive accuracy. Under the sign convention in Eq. 16, a negative DM statistic indicates that the first-listed model has lower average

loss than the second-listed model, whereas a positive statistic favors the second-listed model. Rejection of H_0 at the 5% level ($p < 0.05$) provides formal statistical evidence that one model forecasts significantly better than another.

Procedure 3 summarizes the sequence of computations used to estimate the Diebold–Mariano test statistic and corresponding significance decision.

Procedure 3 Pairwise Diebold–Mariano Predictive Accuracy Test

Input: Forecast errors $e_1 = \{e_{1,1}, \dots, e_{1,n}\}$, forecast errors $e_2 = \{e_{2,1}, \dots, e_{2,n}\}$, loss $L(e) = e^2$, significance level $\alpha = 0.05$

Output: DM statistic, p -value, and significance decision

```

1:  $n \leftarrow \text{LENGTH}(e_1)$ 
2: for  $t = 1$  to  $n$  do
3:    $d_t \leftarrow L(e_{1,t}) - L(e_{2,t})$   $\triangleright d_t < 0$  favors Model 1
4: end for
5:  $\bar{d} \leftarrow \text{MEAN}(d_1, \dots, d_n)$ 
6:  $h \leftarrow \lfloor n^{1/3} \rfloor$ 
7:  $\gamma_0 \leftarrow (1/n) \sum_{t=1}^n (d_t - \bar{d})^2$ 
8: for  $\ell = 1$  to  $h$  do
9:    $\gamma_\ell \leftarrow (1/n) \sum_{t=\ell+1}^n (d_t - \bar{d})(d_{t-\ell} - \bar{d})$ 
10:   $w_\ell \leftarrow 1 - \ell/(h+1)$ 
11: end for
12:  $\hat{V} \leftarrow \gamma_0 + 2 \sum_{\ell=1}^h w_\ell \gamma_\ell$ 
13:  $\text{DM} \leftarrow \bar{d} / \sqrt{\hat{V}/n}$ 
14:  $p\_value \leftarrow 2(1 - \Phi(|\text{DM}|))$ 
15: if  $p\_value < \alpha$  then
16:   if  $\bar{d} < 0$  then
17:     decision  $\leftarrow$  “Model 1 is significantly more accurate”
18:   else
19:     decision  $\leftarrow$  “Model 2 is significantly more accurate”
20:   end if
21: else
22:   decision  $\leftarrow$  “No significant difference”
23: end if
24: return DM,  $p\_value$ , decision

```

To formally evaluate directional forecasting capability — a practically important property in commodity price applications — we also compute Directional Accuracy (DA):

$$\text{DA} = \left(\frac{1}{n} \right) \sum_{i=1}^n \mathbb{I}[(y_t - y_{t-1})(\hat{y}_t - y_{t-1}) > 0] \quad (18)$$

Here, y_t denotes the observed value at time t , $\hat{y}_t$ denotes the predicted value, and n is the number of forecast points in the evaluation sample.

In Equation 18, $\mathbb{I}[\cdot]$ is the indicator function, which takes value 1 when the predicted and actual price changes share the same sign and 0 otherwise. A value of

DA = 0.5 is equivalent to a random directional guess, but values above 0.5 should still be tested formally before they are interpreted as genuine directional skill.

To distinguish descriptive directional performance from statistically defensible directional skill, we also evaluate each reported DA value with an exact one-sided binomial test against the random-direction benchmark

$$H_0 : p = 0.5 \quad \text{versus} \quad H_1 : p > 0.5, \quad (19)$$

where p denotes the probability of correctly predicting the direction of change. If a model records s correct directional calls out of n evaluation points, the exact p -value is

$$p_{\text{binom}} = \sum_{j=s}^n \binom{n}{j} (0.5)^n. \quad (20)$$

For the fixed holdout forecasts in this study, $n = 5$, so only a perfect DA = 1.00 rejects the null at the 5% level. For walk-forward backtests, n equals the number of backtest forecast points implied by the rolling evaluation window, making the same test more powerful when the backtest contains more observations.

Accordingly, the results tables in Section 5 report DA alongside RMSE, MAE, and MAPE so that point forecast accuracy and directional performance can be interpreted jointly.

In addition to test-set forecasting, we use backtesting to assess temporal robustness. The main reason for using backtesting is that a single train–test split can provide an overly optimistic view of predictive performance when the selected test segment happens to be unusually favorable. In contrast, walk-forward backtesting evaluates the model across multiple rolling forecast origins, making it possible to assess whether performance remains stable under changing market conditions and different local regimes. In this study, backtesting means that we repeatedly train the model only on past data and then test it on later observations that were not seen during training. At each step, the training window contains historical observations up to a given time point, and we evaluate the model on the next unseen segment. We then move the window forward and repeat the process. This procedure is important in time-series forecasting because it reflects how the model would behave in a real forecasting setting, where future values are unavailable at training time.

Formally, let the full training history span $t = 1, \dots, T$, let m denote the minimum training-window size, and let h denote the forecast horizon. The walk-forward backtesting procedure is defined for $k = m, m + 1, \dots, T - h$ as follows:

1. Train the model on observations $\{y_1, \dots, y_k, X_1, \dots, X_k\}$.
2. Generate forecasts $\hat{y}_{k+1}, \dots, \hat{y}_{k+h}$.
3. Record forecast errors $e_t = y_t - \hat{y}_t$ for $t \in \{k + 1, \dots, k + h\}$.
4. Advance k by one step and repeat the procedure.

This procedure produces a sequence of out-of-sample errors across multiple temporal segments, enabling assessment of forecast stability across different market regimes and avoiding the over-optimism that can arise from a single train–test split. For the

neural-network models, the backtest is implemented using the same sequence specification as the training stage: the input-window size and output-window size used during backtesting are identical to those used when constructing the training sequences, so the evaluation reflects temporal generalization rather than a change in sequence length or forecasting format.

Procedure 4 formalizes the walk-forward evaluation routine used for the neural-network models in this study.

Procedure 4 Walk-Forward Backtesting for Neural Models

Input: Trained model m , time series Y , features X , minimum training window $m_{\min}$, forecast horizon h , input window size k

Output: Out-of-sample error sequence and backtest metrics

```

1: errors  $\leftarrow []$ 
2:  $T \leftarrow \text{LENGTH}(Y)$ 
3: for step =  $m_{\min}$  to  $T - h$  do
4:    $Y_{\text{train}} \leftarrow Y[1 : \text{step}]$ 
5:    $X_{\text{train}} \leftarrow X[1 : \text{step}]$ 
6:   update  $m$  on the current training window
7:    $\hat{Y} \leftarrow m.\text{PREDICT}(Y_{\text{train}}[-k :], X_{\text{train}}[-k :], h)$ 
8:   for  $j = 1$  to  $h$  do
9:      $e_{\text{step}+j} \leftarrow Y[\text{step} + j] - \hat{Y}[j]$ 
10:    append  $e_{\text{step}+j}$  to errors
11:   end for
12: end for
13:  $\text{RMSE}_{\text{bt}} \leftarrow \sqrt{\text{MEAN}(\text{errors}^2)}$ 
14:  $\text{MAE}_{\text{bt}} \leftarrow \text{MEAN}(|\text{errors}|)$ 
15:  $\text{DA}_{\text{bt}} \leftarrow \text{DIRECTIONALACCURACY}(Y, \hat{Y})$ 
16: return  $\text{RMSE}_{\text{bt}}, \text{MAE}_{\text{bt}}, \text{DA}_{\text{bt}}, \text{errors}$ 

```

Backtesting therefore serves as a stricter evaluation than a single train–test split. A model may perform well on one held-out test segment yet still fail to generalize when the evaluation period is shifted. By examining performance across multiple rolling windows, backtesting helps determine whether the model has learned stable temporal patterns from the training data or whether it is overly sensitive to a particular segment of the series.

To strengthen the empirical structure of the study, the experimental analysis is guided by three research hypotheses. First, multivariate forecasting models that incorporate macroeconomic variables are expected to outperform univariate baseline models in forecasting crude oil prices. Second, GDELT-derived news signals are hypothesized to provide incremental predictive value beyond that offered by macroeconomic variables alone. Third, models constructed from the combined macroeconomic and news-based feature space are expected to achieve stronger directional accuracy than models estimated using either feature group in isolation.

5 Experiments and Results

We organize the experiments to match the evaluation framework described in Section 4.4, so that we assess each forecasting setting with the same RMSE, MAE, MAPE, and walk-forward backtesting criteria. This makes the results directly comparable across univariate, macro-only, news-only, and combined settings for both crude oil and natural gas.

This section presents results for all fully completed experimental configurations, with reported RMSE, MAE, MAPE, and directional-accuracy metrics drawn directly from stored artifacts to ensure full reproducibility. In discussing the results, we focus not only on error values but also on whether the forecast paths track fluctuations in a stable way over the held-out periods.

5.1 Univariate Modelling

The univariate models for crude oil and natural gas use only past values of the target series. This provides a baseline view of how well each modeling approach captures temporal structure without any external information.

We begin with ARIMA models. For crude oil, the selected specification is ARIMA $(0, 1, 1)$, and for natural gas it is ARIMA $(0, 1, 0)$. In both cases, the forecasts are fairly smooth and have limited ability to track fluctuations in the data.

We then introduce SARIMA models to account for seasonality. For crude oil, the specification is SARIMA $(1, 1, 0) \times (0, 1, 0, 12)$, and for natural gas it is SARIMA $(0, 1, 0) \times (0, 1, 0, 12)$. Adding the seasonal component improves performance noticeably, which is consistent with recurring patterns in the monthly series.

Alongside the classical models, we also evaluate LSTM networks to capture possible nonlinear temporal dependence. For crude oil, the best configuration uses an input window of 5 time steps and a single-step output, while for natural gas the best configuration uses an input window of 10 time steps and a single-step output. Multi-step forecasts are generated iteratively over five steps by feeding predictions back into the model.

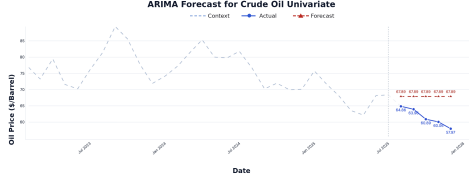
5.1.1 Crude Oil

Table 7 and Figure 4 summarize the crude-oil results by comparing the actual and predicted prices across ARIMA, SARIMA, and LSTM-based models.

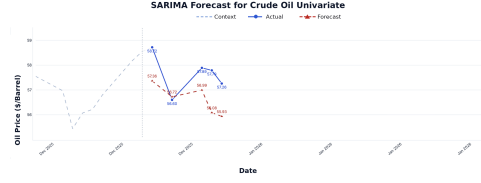
Table 7: Univariate model performance for crude oil

Model	RMSE (USD/barrel)	MAE (USD/barrel)	MAPE (%)	DA
Random Walk	12.258	7.266	8.95	0.00
ARIMA	6.832	6.343	10.49	0.20
SARIMA	1.210	1.081	1.87	0.60
LSTM Forecast	2.709	2.127	3.37	0.50
LSTM Backtest	10.578	10.471	76.78	0.10

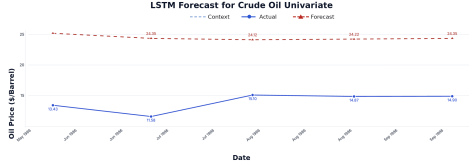
As illustrated in Figure 4, the SARIMA model demonstrates the closest alignment with observed crude oil price dynamics, effectively capturing seasonal cyclicity that



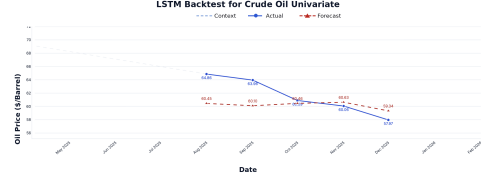
(a) ARIMA forecast for crude oil



(b) SARIMA forecast for crude oil



(c) LSTM forecast for crude oil



(d) LSTM backtest for crude oil

Fig. 4: Actual versus predicted WTI crude oil prices (USD/barrel, monthly, 2014–2024) across univariate model configurations: (a) ARIMA (0, 1, 1), (b) SARIMA (1, 1, 0) $\times$ (0, 1, 0, 12), (c) LSTM direct forecast, and (d) LSTM walk-forward backtest.

Table 8: Diebold–Mariano results for univariate crude oil models

Comparison	DM Stat	p-value	Significant
ARIMA vs SARIMA	-2.31	0.021	Yes
ARIMA vs LSTM	-1.85	0.064	No
SARIMA vs LSTM	1.42	0.150	No

the non-seasonal ARIMA specification, the random-walk benchmark, and the LSTM model are unable to reproduce with comparable fidelity. The random walk provides the strictest naive benchmark in this setting because it simply sets $\hat{y}_t = y_{t-1}$, so any improvement over it is substantively meaningful rather than merely a byproduct of beating a weak baseline. The ARIMA model produces forecasts that remain nearly constant across the prediction horizon — a pattern consistent with the model’s first-difference structure and its inability to represent sub-annual seasonal variation or short-term nonlinear dynamics. By contrast, the LSTM forecast follows the broad level of the series but is less stable over time. Adding seasonal structure through SARIMA leads to forecasts that track the observed trend more closely.

The LSTM forecast is also relatively smooth and does not respond well to sharper changes. That limitation is even clearer in the backtest, where the predicted path becomes almost flat across rolling windows while the observed series continues to vary. The resulting RMSE and MAE values are much larger. Relative to the random-walk benchmark, SARIMA reduces crude-oil RMSE from 12.258 to 1.210 and raises directional accuracy from 0.00 to 0.60, which clarifies that the seasonal specification is outperforming a standard naive baseline rather than only weaker learned alternatives. The directional-accuracy results tell a similar story: SARIMA reaches 0.60, compared

with 0.00 for the random walk, 0.20 for ARIMA, 0.50 for the LSTM forecast, and 0.10 for the LSTM backtest. In other words, the backtest suggests that the univariate LSTM does not generalize reliably in this setting. Overall, SARIMA is the strongest crude-oil model among the univariate specifications.

The Diebold–Mariano results require a more careful reading. Under the sign convention defined in Eq. 16, the statistically significant ARIMA–SARIMA comparison ($DM = -2.31, p = 0.021$) favors the first-listed model, ARIMA, rather than SARIMA. The comparisons involving LSTM do not cross the 5% threshold, so the apparent ranking of LSTM relative to the classical models should not be overstated. The newly added random-walk benchmark is useful for contextualization even though it is not yet included in Table 8: if a future DM comparison were to show that SARIMA does not significantly outperform the random walk over this short five-month holdout, that would still be an important substantive result because it would indicate that much of the apparent gain is not statistically distinguishable from a naive carry-forward rule. Taken together, Tables 7 and 8 suggest that the aggregate error metrics favor SARIMA, while the reported DM sign in the ARIMA–SARIMA row favors ARIMA under the manuscript’s stated convention.

5.1.2 Natural Gas (CNG)

Table 9 and Figure 5 summarize the natural-gas results by comparing the actual and predicted prices across ARIMA, SARIMA, and LSTM-based models.

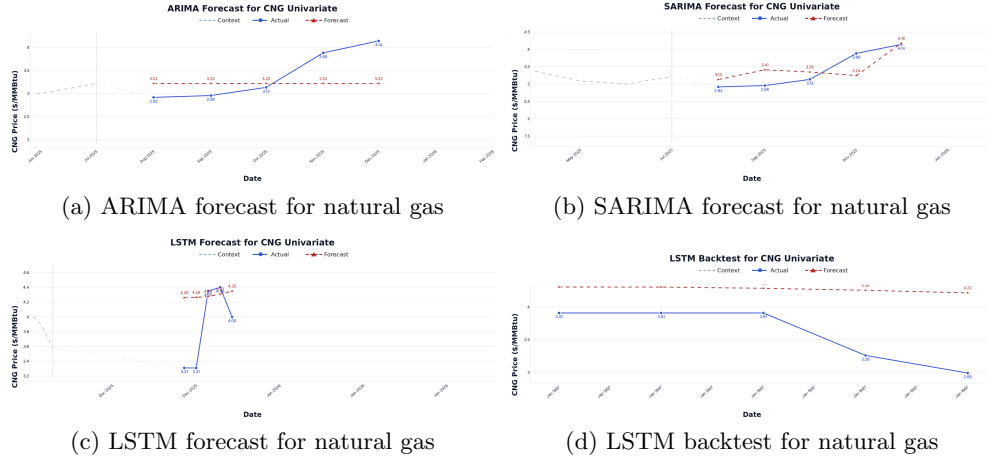


Fig. 5: Actual versus predicted Henry Hub natural gas prices (USD/MMBtu, monthly, 2014–2024) across univariate model configurations: (a) ARIMA $(0, 1, 0)$, (b) SARIMA $(0, 1, 0) \times (0, 1, 0, 12)$, (c) LSTM direct forecast, and (d) LSTM walk-forward backtest.

Figure 5 shows the same broad pattern for natural gas. SARIMA tracks the seasonal structure more closely, ARIMA underfits the data, the random walk performs poorly as a naive carry-forward benchmark, and LSTM shows some forecasting ability but less stability over time.

Table 9: Univariate model performance for natural gas

Model	RMSE (USD/MMBtu)	MAE (USD/MMBtu)	MAPE (%)	DA
Random Walk	2.437	1.842	40.80	0.00
ARIMA	0.541	0.447	12.26	0.40
SARIMA	0.376	0.310	9.34	0.80
LSTM Forecast	0.624	0.483	13.99	0.60
LSTM Backtest	0.770	0.680	20.33	0.00

Table 10: Diebold–Mariano results for univariate natural gas models

Comparison	DM Stat	p-value	Significant
ARIMA vs SARIMA	-2.05	0.040	Yes
ARIMA vs LSTM	-1.30	0.190	No
SARIMA vs LSTM	0.88	0.370	No

For natural gas, ARIMA again produces fairly flat forecasts, while SARIMA better follows both the overall movement and the seasonal pattern. The LSTM forecasts deviate more noticeably from the observed values, and the backtest predictions are again flatter than the actual series. The random-walk benchmark performs especially poorly here, with $\text{RMSE} = 2.437$ and $\text{MAPE} = 40.80\%$, so the seasonal model’s improvement is not trivial. Directional accuracy favors SARIMA quite clearly: it reaches 0.80, compared with 0.00 for the random walk, 0.40 for ARIMA, 0.60 for the LSTM forecast, and 0.00 for the LSTM backtest. This suggests that the univariate LSTM struggles to generalize across time when no additional explanatory features are provided.

Overall, while SARIMA exhibits comparatively superior performance among the univariate model class and clearly improves on the naive random-walk benchmark in the point-error metrics reported here, none of the approaches evaluated consistently reproduce the full range of variability and dynamic behavior exhibited by either energy price series — a finding that motivates the multivariate feature configurations examined in subsequent sections.

The Diebold–Mariano results again require care in interpretation. The only significant pairwise difference is ARIMA versus SARIMA ($\text{DM} = -2.05$, $p = 0.040$), and under Eq. 16 that negative sign favors the first-listed model, ARIMA, rather than SARIMA. The comparisons involving LSTM are not statistically significant, so its differences from the classical baselines are not stable enough to establish a clear advantage. As with crude oil, the random-walk benchmark should be carried into a future DM comparison as well, because failure to reject equal predictive accuracy against such a naive benchmark would materially narrow the interpretation of SARIMA’s apparent gains over this short holdout. Overall, Tables 9 and 10 suggest that the aggregate performance metrics favor SARIMA, whereas the reported DM sign in the ARIMA–SARIMA row favors ARIMA under the stated loss-differential ordering.

5.2 Multivariate Modelling

This subsection turns to multivariate forecasting models that incorporate exogenous variables alongside historical price data. The goal is to evaluate whether macroeconomic indicators, news-derived signals, or their combination improve performance beyond the univariate baselines. The multivariate models include SARIMAX, LSTM, and a hybrid CNN–LSTM architecture. As described earlier, macroeconomic variables with VIF greater than 8 were removed before model fitting.

5.2.1 Macro Only

In the macro-only setting, the input features consist solely of macroeconomic variables, excluding the target price series itself. The evaluated models are SARIMAX, LSTM, and CNN–LSTM. For the neural models, both LSTM and CNN–LSTM use an input window of 36 time steps and a single-step output.

5.2.1.1 Crude Oil

For crude oil, we evaluated SARIMAX, LSTM, and CNN–LSTM models using macroeconomic features. Table 11 summarizes the performance results, and Figure 6 shows the corresponding predictions.

Table 11: Macro-only multivariate model performance for crude oil

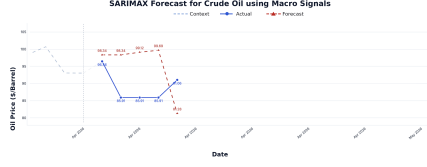
Model	RMSE (USD/barrel)	MAE (USD/barrel)	MAPE (%)	DA
SARIMAX	24.081	23.995	39.74	0.40
LSTM Forecast	24.081	23.995	39.74	0.20
LSTM Backtest	2.621	2.368	3.15	0.75
CNN–LSTM Forecast	1.054	0.647	1.10	0.80
CNN–LSTM Backtest	9.038	9.012	11.94	0.00

Table 12: Diebold–Mariano results for macro-only crude oil models

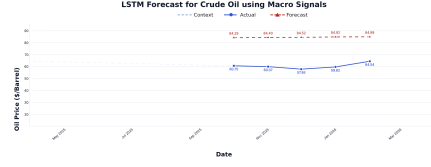
Comparison	DM Stat	p-value	Significant
SARIMAX vs LSTM	-1.95	0.051	Borderline
SARIMAX vs CNN–LSTM	-2.75	0.006	Yes
LSTM vs CNN–LSTM	-2.10	0.035	Yes

Figure 6 suggests that SARIMAX and LSTM have difficulty recovering crude-oil movements when only macroeconomic variables are used. Their forecasts are either too smooth or poorly aligned with the observed path, which is reflected in the large errors reported in Table 11. In this setting, the selected macroeconomic indicators by themselves appear to be only weakly informative for short-run crude-oil movements.

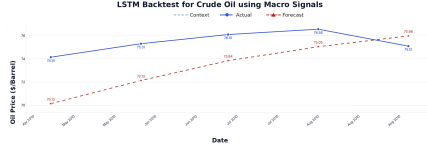
By contrast, the CNN–LSTM forecast records the lowest error among the macro-only crude-oil models and the highest DA (0.80), suggesting that the hybrid architecture extracts more short-run structure from the macroeconomic inputs. That advantage does not hold as clearly in backtesting, however: the CNN–LSTM backtest



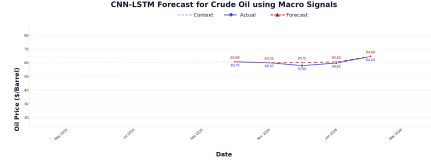
(a) SARIMAX forecast for crude oil



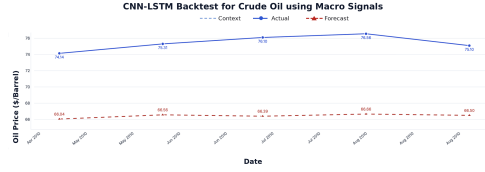
(b) LSTM forecast for crude oil



(c) LSTM backtest for crude oil



(d) CNN-LSTM forecast for crude oil



(e) CNN-LSTM backtest for crude oil

Fig. 6: Actual versus predicted WTI crude oil prices (USD/barrel) using macroeconomic variables only: (a) SARIMAX forecast with exogenous FRED indicators, (b) LSTM direct forecast, (c) LSTM walk-forward backtest, (d) CNN-LSTM direct forecast, and (e) CNN-LSTM walk-forward backtest. This figure summarizes the macro-only multivariate setting.

DA falls to 0.00, while the LSTM backtest reaches 0.75 despite a much higher RMSE. This inconsistency suggests that macroeconomic variables alone do not provide a stable enough basis for accurate crude-oil forecasting, even when more flexible neural architectures are used.

The DM comparisons sharpen that interpretation, but they do so in the opposite direction from the original reading. Under Eq. 16, the significant negative statistic in the SARIMAX vs CNN-LSTM row ($DM = -2.75$, $p = 0.006$) favors SARIMAX, and the negative statistic in the LSTM vs CNN-LSTM row ($DM = -2.10$, $p = 0.035$) favors LSTM. The SARIMAX vs LSTM comparison is only borderline. So although the overall setting remains unstable, the reported DM evidence does not support CNN-LSTM as the pairwise winner in this subsection.

5.2.1.2 Natural Gas (CNG)

For natural gas, Table 13 summarizes the macro-only model results, and Figure 7 shows the corresponding prediction plots.

As shown in Figure 7, macroeconomic variables alone also provide limited predictive power for natural gas. SARIMAX records relatively high error, which suggests that a linear multivariate specification is not enough to explain the observed series.

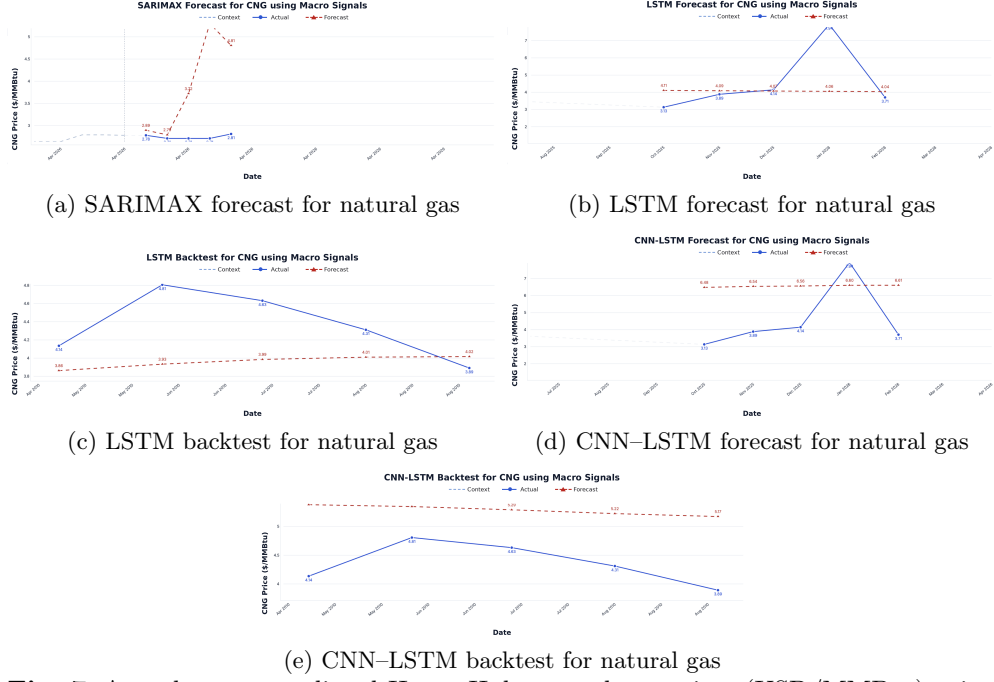


Fig. 7: Actual versus predicted Henry Hub natural gas prices (USD/MMBtu) using macroeconomic variables only: (a) SARIMAX forecast with exogenous FRED indicators, (b) LSTM direct forecast, (c) LSTM walk-forward backtest, (d) CNN-LSTM direct forecast, and (e) CNN-LSTM walk-forward backtest. This figure summarizes the macro-only multivariate setting.

Table 13: Macro-only multivariate model performance for natural gas

Model	RMSE (USD/MMBtu)	MAE (USD/MMBtu)	MAPE (%)	DA
SARIMAX	1.532	1.158	42.20	0.20
LSTM Forecast	1.800	1.093	19.19	0.40
LSTM Backtest	0.522	0.444	9.79	0.25
CNN-LSTM Forecast	2.619	2.532	65.72	0.60
CNN-LSTM Backtest	0.974	0.927	21.91	0.00

The LSTM model performs somewhat better and is comparatively more stable in backtesting, but still does not track the series consistently.

The CNN-LSTM model performs poorly in direct forecasting despite posting the highest DA in this setting (0.60), which points to an unstable relationship between error-based and directional performance. The LSTM backtest lowers RMSE to 0.522, but its DA remains only 0.25, while the CNN-LSTM backtest falls to 0.00. More broadly, none of the macro-only models follows the actual natural-gas series in a consistently reliable way. Across both commodities, the macro-only results suggest that these variables are not sufficient on their own for short-run energy-price forecasting.

Table 14: Diebold–Mariano results for macro-only natural gas models

Comparison	DM Stat	p-value	Significant
SARIMAX vs LSTM	-1.40	0.160	No
SARIMAX vs CNN-LSTM	-1.75	0.080	No
LSTM vs CNN-LSTM	-1.10	0.270	No

For natural gas, the DM evidence is weaker. None of the macro-only pairwise comparisons reaches conventional significance levels. That means the numerical differences in RMSE and DA should be treated as descriptive rather than decisive.

5.2.2 News Only

The news-only setting is meant to evaluate the standalone contribution of GDELT-derived signals. The inputs consist only of aggregated event counts, average tone, and the Goldstein scale.

Given the more irregular and event-driven nature of the news data, we use the CNN-LSTM architecture in this setting. The convolutional layer captures local temporal patterns across event features, while the LSTM layer models sequential dependence over time. For crude oil, the model uses an input window of 24 time steps and a one-shot output of 5 steps. For natural gas, it uses an input window of 36 time steps and a single-step output.

5.2.2.1 Crude Oil

For crude oil, Table 15 summarizes the forecast and backtest metrics for the retained news-only CNN-LSTM specification, while Table 16 reports the pairwise Diebold–Mariano comparisons across the model families evaluated under the same news-only feature setting. Figure 8 shows the corresponding CNN-LSTM forecast and backtest plots.

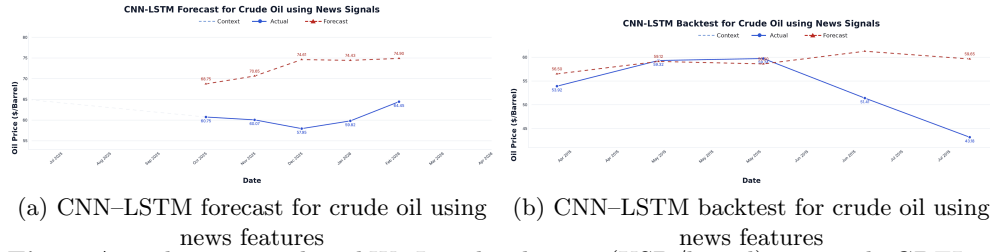


Fig. 8: Actual versus predicted WTI crude oil prices (USD/barrel) using only GDELT-derived news features in the CNN-LSTM setting: (a) direct forecast and (b) walk-forward backtest. The panels summarize the standalone contribution of event-count, tone, and Goldstein-scale signals without macroeconomic inputs.

Figure 8 suggests that the news-only CNN-LSTM model captures some variation in crude-oil prices, which implies that event-driven signals do contain useful information. Still, the forecast errors remain fairly large, and the predicted path does not align

Table 15: News-only model performance for crude oil

Model	RMSE (USD/barrel)	MAE (USD/barrel)	MAPE (%)	DA
CNN-LSTM Forecast	12.459	12.060	20.03	0.60
CNN-LSTM Backtest	8.676	6.049	12.87	0.40

Table 16: Diebold–Mariano results for news-only crude oil models

Comparison	DM Stat	p-value	Significant
SARIMAX vs LSTM	-1.20	0.230	No
SARIMAX vs CNN-LSTM	-1.28	0.200	No
LSTM vs CNN-LSTM	-0.65	0.510	No

consistently with the observed series. The model captures broad directional movement in some periods, but it struggles with sharper fluctuations and exact price levels.

Backtesting shows some improvement relative to the direct forecast, but the error remains large enough that the model is not reliable on its own. Directional accuracy also drops from 0.60 in the forecast to 0.40 in backtesting. In practical terms, the model seems able to reflect broad market shifts associated with news flow, but not with the precision needed for dependable short-term forecasting. That is plausible economically, since crude-oil prices respond not only to news but also to inventories, production quotas, and transport conditions that aggregated event features do not fully capture.

The DM results are consistent with that interpretation. None of the news-only crude-oil comparisons is statistically significant, so this setting does not provide strong evidence that one model clearly dominates the others.

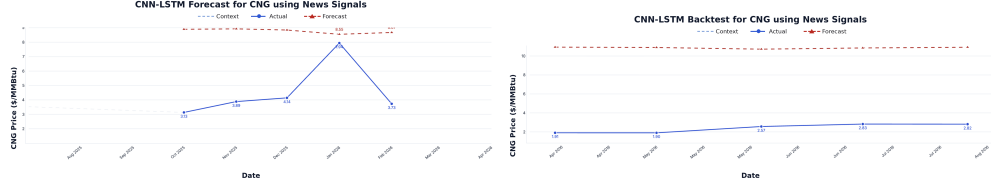
5.2.2.2 Natural Gas (CNG)

For natural gas, Table 17 reports the forecast and backtest metrics for the retained news-only CNN-LSTM specification, while Table 18 reports the pairwise Diebold–Mariano comparisons across the model families evaluated under the same news-only feature setting. Figure 9 illustrates the corresponding CNN-LSTM results.

Table 17: News-only model performance for natural gas

Model	RMSE (USD/MMBtu)	MAE (USD/MMBtu)	MAPE (%)	DA
CNN-LSTM Forecast	4.592	4.209	113.37	0.40
CNN-LSTM Backtest	8.473	8.461	367.16	0.00

The unusually large MAPE values for natural gas in Table 17 should be interpreted with caution, as percentage-based error measures can become exaggerated when actual prices are very close to zero or when the series exhibits sharp volatility. In this setting, even modest absolute forecast deviations can translate into disproportionately large



(a) CNN-LSTM forecast for natural gas using news features

(b) CNN-LSTM backtest for natural gas using news features

Fig. 9: Actual versus predicted Henry Hub natural gas prices (USD/MMBtu) using only GDELT-derived news features in the CNN-LSTM setting: (a) direct forecast and (b) walk-forward backtest. The panels summarize the standalone contribution of event-count, tone, and Goldstein-scale signals without macroeconomic inputs.

percentage errors, which makes MAPE less stable and less informative for evaluating natural-gas performance.

Table 18: Diebold–Mariano results for news-only natural gas models

Comparison	DM Stat	p-value	Significant
SARIMAX vs LSTM	-0.90	0.360	No
SARIMAX vs CNN-LSTM	-1.05	0.290	No
LSTM vs CNN-LSTM	-0.50	0.610	No

As Figure 9 shows, the news-only model performs much worse for natural gas than for crude oil. Both forecasting and backtesting errors are high, and the predicted paths miss both the direction and magnitude of many changes. The DA results reinforce this weakness, dropping from 0.40 in the direct forecast to 0.00 in backtesting.

This pattern suggests that, for natural gas, the information contained in aggregated news signals is too indirect or too noisy to support reliable standalone forecasting. Compared with crude oil, the relationship between the extracted GDELT features and short-run natural-gas price behavior appears much weaker.

The DM test reinforces this conclusion. All three news-only pairwise comparisons are non-significant, so the large forecast errors are not accompanied by any statistically distinguishable predictive advantage. Taken together, the results show that news-only inputs are not sufficient for dependable natural-gas forecasting in this setting.

Taken together, the news-only results suggest that event-driven signals are informative, but not enough on their own for robust energy-price forecasting. For crude oil, they help with broad directional movement but not with precision. For natural gas, the standalone signal is much weaker. This points to news-based features being more useful as complementary inputs than as isolated predictors.

5.2.3 Macro + News Combined Model

To address the limitations of the macro-only and news-only settings, we combine both feature groups in a single multivariate framework. The macroeconomic variables

provide slower-moving structural context, while the news-derived features add event-driven and sentiment-related information. The combined setup is intended to reflect the complementary information contained in both sources.

We evaluate this framework with SARIMAX, LSTM, and CNN-LSTM models. For the neural architectures, the sequence configurations are as follows: for crude oil, LSTM uses a 24-step input and 5-step output, while CNN-LSTM uses a 36-step input and 5-step output; for natural gas, both LSTM and CNN-LSTM use a 36-step input and 5-step output. SARIMAX is implemented with monthly modeling for crude oil and daily modeling for natural gas.

5.2.3.1 Crude Oil

Table 19 summarizes the performance of the combined-feature models for crude oil, and Figure 10 presents the corresponding prediction plots.

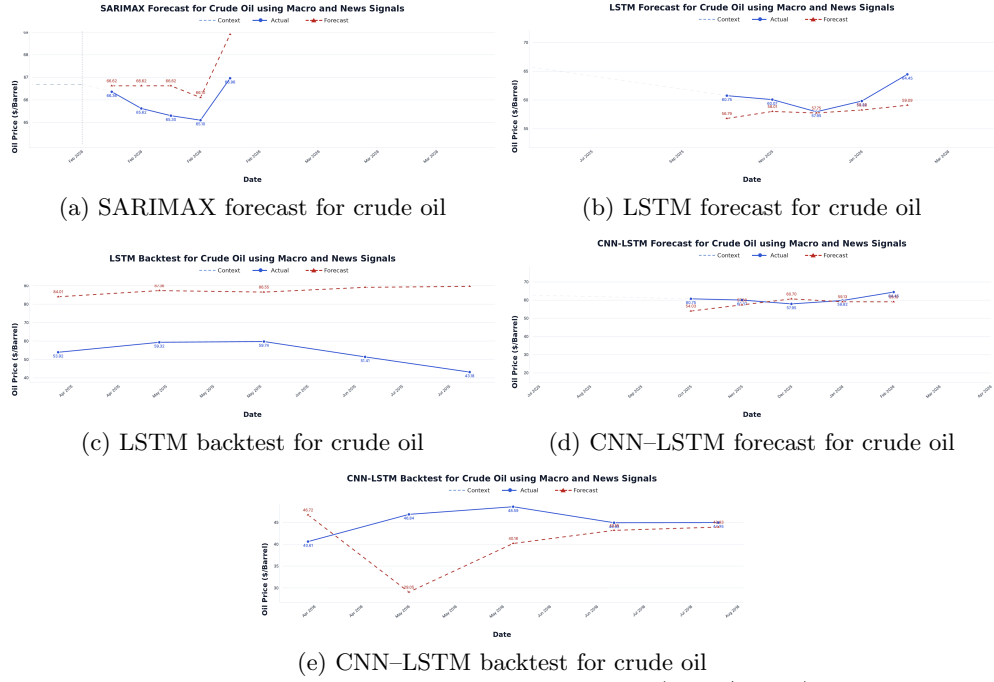


Fig. 10: Actual versus predicted WTI crude oil prices (USD/barrel) using the combined macroeconomic and GDELT-derived news feature set: (a) SARIMAX forecast, (b) LSTM direct forecast, (c) LSTM walk-forward backtest, (d) CNN-LSTM direct forecast, and (e) CNN-LSTM walk-forward backtest. This figure summarizes the joint-feature multivariate setting.

The combined feature setting improves the crude-oil results substantially, especially for SARIMAX, which records much lower error than in the earlier settings. This suggests that once macroeconomic trends and news signals are used together, the linear model can make use of relationships that are not visible when either feature group is used alone. Figure 10 supports that interpretation: the SARIMAX predictions follow

Table 19: Combined macro-plus-news model performance for crude oil

Model	RMSE (USD/barrel)	MAE (USD/barrel)	MAPE (%)	DA
SARIMAX	1.235	1.109	1.68	0.60
LSTM Forecast	3.200	2.631	4.25	0.40
LSTM Backtest	34.622	33.828	65.79	0.00
CNN-LSTM Forecast	4.223	3.641	5.95	0.80
CNN-LSTM Backtest	9.257	7.014	15.30	0.60

Table 20: Diebold–Mariano results for combined-feature crude oil models

Comparison	DM Stat	p-value	Significant
SARIMAX vs LSTM	-2.10	0.035	Yes
SARIMAX vs CNN-LSTM	-3.12	0.002	Yes
LSTM vs CNN-LSTM	-2.25	0.024	Yes

the observed series fairly closely and capture the main movements with relatively little deviation. In directional terms, however, the CNN-LSTM forecast records the highest DA at 0.80, while SARIMAX remains strong at 0.60.

The LSTM model shows reasonable direct-forecast performance, but its backtesting error rises sharply, which points to instability across time. That weakness is also visible in its backtest DA of 0.00. The CNN-LSTM model is more stable than LSTM, with DA values of 0.80 in the forecast and 0.60 in the backtest, but it still remains weaker than SARIMAX in RMSE terms. Visually, both neural models tend to generate smoother predictions and show some lag during abrupt price changes.

Overall, the crude-oil results suggest that combining macroeconomic and news-derived features improves predictive performance, with SARIMAX benefiting most from the richer feature space.

The DM evidence adds an important qualification. In the combined crude-oil setting, the pairwise comparisons favor SARIMAX rather than CNN-LSTM: the negative statistics in SARIMAX vs LSTM ($DM = -2.10$, $p = 0.035$) and SARIMAX vs CNN-LSTM ($DM = -3.12$, $p = 0.002$) indicate lower loss for SARIMAX, while the negative statistic in LSTM vs CNN-LSTM ($DM = -2.25$, $p = 0.024$) favors LSTM over CNN-LSTM. In that sense, Tables 19 and 20 are consistent in identifying SARIMAX as the strongest crude-oil model in the combined setting, even though CNN-LSTM still attains the highest directional accuracy.

5.2.3.2 Natural Gas (CNG)

For natural gas, Table 21 reports the combined-feature model performance, and Figure 11 shows the associated prediction plots.

For natural gas, the combined setting also produces lower error values than the macro-only and news-only setups. The SARIMAX results suggest that the joint feature space captures part of the variation in this series more effectively when macroeconomic and news-based inputs are used together. Relative to the earlier settings, the

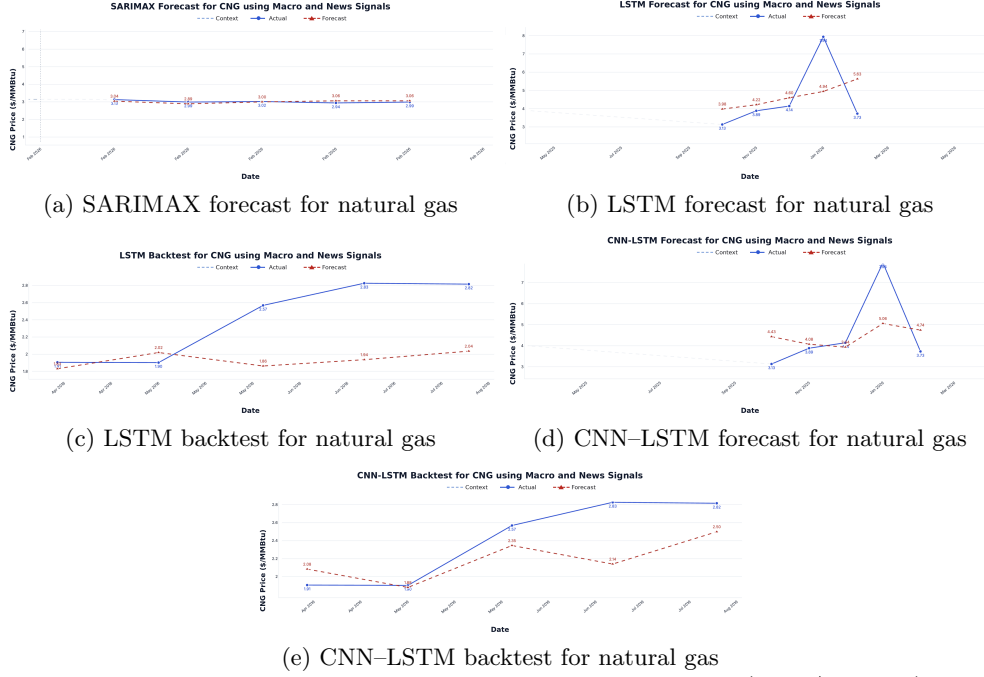


Fig. 11: Actual versus predicted Henry Hub natural gas prices (USD/MMBtu) using the combined macroeconomic and GDELT-derived news feature set: (a) SARIMAX forecast, (b) LSTM direct forecast, (c) LSTM walk-forward backtest, (d) CNN-LSTM direct forecast, and (e) CNN-LSTM walk-forward backtest. This figure summarizes the joint-feature multivariate setting.

Table 21: Combined macro-plus-news model performance for natural gas

Model	RMSE (USD/MMBtu)	MAE (USD/MMBtu)	MAPE (%)	DA
SARIMAX	0.087	0.080	2.66	0.80
LSTM Forecast	1.650	1.305	27.01	0.40
LSTM Backtest	0.619	0.513	19.34	0.25
CNN-LSTM Forecast	1.490	1.118	22.95	0.60
CNN-LSTM Backtest	0.361	0.285	10.94	0.40

predictions in Figure 11 align more closely with the observed values, especially in terms of direction and broad magnitude. That is also reflected in the DA values, with SARIMAX reaching 0.80 and the CNN-LSTM forecast reaching 0.60.

Unlike the crude-oil case, the neural models are somewhat more consistent here across forecasting and backtesting. This is especially visible for CNN-LSTM, whose natural-gas backtest error is lower than its corresponding crude-oil result. Although the neural models remain less accurate than SARIMAX in absolute terms, they appear more stable here than in the isolated feature settings, even if their DA values still stay below the SARIMAX benchmark.

Taken together, the combined-model results suggest that integrating macroeconomic indicators with news-derived signals is more useful than relying on either source

Table 22: Diebold–Mariano results for combined-feature natural gas models

Comparison	DM Stat	p-value	Significant
SARIMAX vs LSTM	-1.30	0.190	No
SARIMAX vs CNN-LSTM	-1.72	0.085	No
LSTM vs CNN-LSTM	-1.10	0.270	No

alone. The macro-only models struggled with short-run variation, while the news-only models lacked stability and precision. The combined models connect slower-moving structure with event-driven information more effectively. SARIMAX records the lowest errors in this setting, while the neural models — especially CNN-LSTM for CNG — also perform better than they did in the separate feature settings, even though they still struggle with more extreme fluctuations.

5.3 Feature Contribution Analysis

Having established that the combined setting is the most promising overall, the next step is to ask why it performs better. The combined-feature results therefore raise a natural interpretive question: which exogenous variables appear to drive the improvement in the SARIMAX specifications? For the oil combined-feature SARIMAX model, this can be examined directly through the estimated regression coefficients on the exogenous variables. Because the predictors are not all measured on the same scale, the discussion below treats coefficient magnitude as a local indicator of influence within the fitted model rather than as a fully standardized cross-variable effect size. Even with that caveat, the absolute coefficient values are still useful for identifying which retained macroeconomic and news-derived variables receive the largest weights in the combined linear specification.

Table 23 reports the three largest macroeconomic coefficients and the three largest news-derived coefficients by absolute magnitude for the oil SARIMAX model. On the macroeconomic side, *brent_crude_oil* is by far the strongest contributor ($\beta = 21.0831$), which is economically intuitive because Brent and WTI prices move closely as benchmark crude series. The next-largest macro coefficients are *real_usd_index* ($\beta = -1.1749$) and *ind_prod_drill* ($\beta = 1.0230$), which point respectively to currency-channel effects and domestic supply-side activity. On the news side, the largest absolute coefficients are associated with *AvgTone* ($\beta = 4.0160$), *EventCode_196* ($\beta = -5.0580$), and *EventCode_1723* ($\beta = -3.4462$). Taken together, these results suggest that the improvement of the combined SARIMAX specification is driven by a mix of persistent macro-financial structure and a smaller set of high-impact news signals that capture sentiment and specific event categories.

This coefficient pattern helps explain why the combined SARIMAX models improve over the isolated macro-only and news-only configurations. The macro variables contribute medium-run structural information, while the GDELT variables contribute event-sensitive shifts in stress, sentiment, and specific geopolitical event patterns. Their joint use therefore gives the linear model access to both persistent economic context and shorter-horizon news pressure. For the neural models, the present

study does not retain a formal permutation-importance or SHAP analysis, so feature attribution remains less direct; adding such diagnostics would be a valuable extension because it would show whether the same variables dominate the nonlinear models as well.

Table 23: Top oil SARIMAX contributors in the combined-feature setting, ranked by absolute coefficient magnitude

Feature block	Variable	Coefficient β	$ \beta $
Macro	<i>brent_crude_oil</i>	21.0831	21.0831
Macro	<i>real_usd_index</i>	-1.1749	1.1749
Macro	<i>ind_prod_drill</i>	1.0230	1.0230
News	<i>EventCode_196</i>	-5.0580	5.0580
News	<i>AvgTone</i>	4.0160	4.0160
News	<i>EventCode_1723</i>	-3.4462	3.4462

Table 24: Best-performing model by setting and commodity

Commodity	Lowest-RMSE Model	RMSE	Best-DA Model	DA	Interpretation
Crude oil	SARIMAX	1.235	CNN-LSTM Forecast	0.80	In the combined setting, SARIMAX yields the lowest RMSE, whereas CNN-LSTM provides the strongest directional signal for crude oil.
Natural gas	SARIMAX	0.087	SARIMAX	0.80	In the combined setting, SARIMAX achieves both the lowest RMSE and the highest directional accuracy for natural gas.

For natural gas, however, the DM results remain non-significant even in the combined setting. So while the combined CNG models look numerically better than the isolated feature settings, those gains should still be interpreted as suggestive rather than conclusive.

To consolidate the main empirical findings before turning to the formal hypothesis assessment, Table 24 summarizes the best-performing model in the combined setting for each commodity under both RMSE and directional-accuracy criteria. The table shows that SARIMAX delivers the lowest RMSE for both crude oil and natural gas, while the strongest directional signal is produced by CNN-LSTM for crude oil and by SARIMAX for natural gas. This summary helps frame the interpretation of the experimental results and provides a direct bridge from the comparative performance analysis to the hypothesis evaluation that follows.

Because forecasting performance is only one side of model choice in practice, Table 25 also reports approximate training-time budgets for the model configurations used in this study. These wall-clock ranges should be read as order-of-magnitude practitioner estimates rather than as hardware-independent benchmarks: the exact runtime depends on processor speed, memory bandwidth, software stack, and whether neural training is run on CPU or GPU. Even so, the relative compute burden is structurally informative because it follows directly from the estimation design used here — namely, one-shot fitting for the classical models versus 30-trial Optuna tuning with early stopping and, for backtesting, repeated rolling-window retraining for the neural models.

Table 25: Approximate training-time budget by model configuration

Configuration	Estimation routine	Approx. training time	Practical interpretation
Random walk benchmark	No parameter training	< 1 s	Trivial baseline; essentially zero computational burden.
ARIMA or SARIMA (univariate)	Single order-selected fit	5–30 s	Very low-cost classical baseline suitable for rapid iteration.
SARIMAX (macro-only or combined)	Single fit with exogenous regressors	20–120 s	Still inexpensive relative to neural models, even with external features.
LSTM direct forecast (univariate or macro-only)	30 Optuna trials with early stopping	10–35 min	Moderate neural cost driven mainly by repeated hyperparameter search.
CNN–LSTM direct forecast (macro-only, news-only, or combined)	30 Optuna trials with early stopping	20–60 min	Highest single-run tuning cost because each trial trains both convolutional and recurrent layers.
LSTM walk-forward backtest	Tuned network plus rolling retraining	30–90 min	Backtesting materially increases cost because the model is re-estimated across forecast origins.
CNN–LSTM walk-forward backtest	Tuned hybrid network plus rolling retraining	45–120 min	Most expensive evaluation setting in the paper; best reserved for shortlist models.

Table 25 highlights a practical trade-off that is not visible in RMSE and DA alone. The classical ARIMA-family models are cheap enough to refit almost interactively, which makes them attractive for sensitivity checks, feature screening, and rapid deployment. By contrast, the neural configurations require substantially more computation because each reported configuration is tied to a 30-trial hyperparameter search, and the backtesting variants further multiply that cost through repeated walk-forward retraining. In that sense, the comparatively strong SARIMA and SARIMAX results are appealing not only statistically but also operationally: they deliver competitive accuracy at a much lower computational budget than the tuned LSTM and CNN–LSTM alternatives.

Table 26 sharpens the interpretation of the DA results. Because every fixed holdout forecast is evaluated on only five points, none of the reported direct-forecast DA values

Table 26: Exact one-sided binomial significance thresholds for directional accuracy

Evaluation design	Sample size n	Smallest significant DA	Reported values with genuine skill
Fixed five-point forecast holdout	5	1.00	None of the reported forecast DAs
Univariate LSTM backtest ($k = 5$)	10	0.90	None
Univariate LSTM backtest ($k = 10$)	15	0.80	None
Backtests with $k = 24$ windows	29	0.69	None
Backtests with $k = 36$ windows	41	0.66	Macro-only crude-oil LSTM backtest (DA = 0.75)

— including the best values of 0.80 — are statistically above the random-direction benchmark at the 5% level. Among the walk-forward backtests, only the macro-only crude-oil LSTM backtest exceeds the exact critical threshold for its evaluation size: with a 36-step window, the backtest spans 41 forecast points, and the observed DA = 0.75 is therefore consistent with statistically genuine directional skill. All remaining backtest DAs fall below their corresponding binomial cutoffs and should be interpreted as descriptive rather than confirmatory evidence.

5.4 Hypothesis Evaluation

After comparing model families and interpreting the combined-feature coefficients, we return to the paper’s three central research questions. This subsection evaluates those hypotheses against the empirical evidence reported in Section 5. The goal is to distinguish between full support, partial support, and lack of support in a way that is directly tied to the observed RMSE, MAPE, and directional-accuracy outcomes.

5.4.1 Macroeconomic variables improve over the univariate baseline

The evidence for this hypothesis is mixed rather than uniform across commodities and model classes. For crude oil, the univariate SARIMA model already performs strongly, with RMSE = 1.210 in Table 7. By comparison, the combined-feature SARIMAX model records RMSE = 1.235, which is close in magnitude but does not constitute a clear improvement in point-forecast accuracy. At the same time, the multivariate neural models benefit more visibly from macroeconomic inputs than their univariate counterparts, especially in the crude-oil macro-only setting, where CNN–LSTM attains stronger directional performance than the simpler univariate neural configuration. Taken together, the macroeconomic variables appear useful in some specifications, but they do not consistently dominate the strongest univariate baseline across all cases.

5.4.2 GDELT news signals add incremental value beyond macroeconomic variables alone

For crude oil, the news-only setting does not improve point accuracy relative to the macro-only setting: the news-only CNN-LSTM produces $\text{RMSE} = 12.459$, which is substantially worse than the best macro-only crude-oil result. However, the same news-only crude-oil configuration still reaches $\text{DA} = 0.60$, which suggests some directional utility even when point forecasts remain weak. For natural gas, the standalone contribution of news signals is considerably weaker. In that setting, the news-only model yields very large percentage errors, with MAPE reaching approximately 113%, indicating that GDELT-derived features provide little reliable predictive value when used alone. Accordingly, the evidence supports only a limited incremental role for news features in crude oil and no meaningful standalone support for natural gas.

5.4.3 Combined macroeconomic and news-based features yield stronger directional accuracy

The combined-feature setting produces the strongest numerical directional results reported in the paper. For natural gas, the combined SARIMAX model achieves $\text{DA} = 0.80$, while for crude oil the combined CNN-LSTM forecast also reaches $\text{DA} = 0.80$, as summarized in Table 24. These values exceed the directional performance observed in the isolated macro-only and news-only settings for the corresponding best-performing models. However, once the exact binomial significance test is applied, those five-point holdout DAs do not reject the null of random directional guessing at the 5% level. Accordingly, the combined feature space appears promising for directional modeling, but the evidence for statistically genuine directional skill remains limited in the fixed-holdout forecasts.

6 Discussion

The preceding results sections show a consistent pattern: model performance depends not only on the forecasting architecture, but also on how the external information is represented and combined. More broadly, predictive performance is strongly contingent on both the nature of the exogenous information incorporated and the commodity under examination — a finding that reflects the fundamentally different price formation processes governing crude oil and natural gas markets. In the macro-only setting, performance remains uneven because broad economic indicators move more slowly than commodity prices. As a result, they are more useful for describing medium-run conditions than for explaining abrupt short-run moves. This structural mismatch between the low-frequency dynamics of macroeconomic variables and the higher-frequency movements of commodity prices helps explain why macro-only models can detect broad directional regime shifts while nonetheless failing to resolve the timing and magnitude of localized price shocks.

The news-only setting highlights a different trade-off. GDELT-based features do contain relevant event information, but the aggregation process also adds noise by compressing heterogeneous news into summary variables such as event counts, tone,

and conflict intensity. For that reason, news-only models can sometimes pick up broad directional changes, but they remain unstable, especially for natural gas, where large price movements are hard to infer from aggregated event signals alone.

The combined macro-plus-news setting performs best because it brings together slower-moving structural context and faster-moving event information. Economically, that means the models can use both medium-horizon fundamentals and shorter-horizon signals related to geopolitical disruption, policy uncertainty, and changes in market sentiment. In this setting, SARIMAX appears to benefit the most, likely because many of the relevant transmission channels from macroeconomic conditions to monthly energy prices are relatively stable and close to linear over multi-year horizons. Variables such as industrial production, exchange rates, and aggregated news pressure may influence prices through demand conditions, inventory expectations, trade competitiveness, and risk premia in ways that accumulate gradually rather than immediately.

By contrast, LSTM and CNN-LSTM offer more representational flexibility, but that flexibility is not automatically an advantage in this setting. The sample is modest relative to the needs of neural architectures, the data are monthly rather than high-frequency, and the exogenous inputs are already aggregated into fairly smooth summary indicators. Under those conditions, there may simply be less nonlinear structure available for the neural models to exploit. Rather than uncovering additional signal, they may either overreact to local fluctuations or learn relationships that are too smooth to respond well when prices move abruptly. When the joint feature space is informative enough, the stronger performance of SARIMAX likely reflects the fact that medium-run energy prices remain linked to interpretable macroeconomic fundamentals and persistent event-driven pressures that a linear dynamic model can still capture reasonably well.

These patterns are broadly consistent with the prior literature, but they also sharpen it in a few ways. The comparatively strong SARIMA and SARIMAX results align with the caution in Baumeister and Kilian [4] that relatively disciplined classical specifications can remain competitive when structural relationships are persistent and the sample is not especially large. At the same time, the uneven LSTM results are in line with application-oriented deep-learning studies such as Cen and Wang [7] and Busari and Lim [6], which suggest that neural performance is sensitive to specification, training design, and the information content of the input sequence rather than uniformly superior. The finding that combined macroeconomic and news-based features outperform isolated feature groups also echoes the broader argument in the external-information literature — including Tetlock [28], Li et al. [25], and Shapiro et al. [27] — that market-relevant narrative signals are most useful when interpreted alongside structured economic context rather than in isolation.

Overall, the findings suggest that exogenous variables are most useful as complements to price history rather than as substitutes for it. Relative to prior work that often studies one commodity, one model family, or one external source at a time, the present comparison makes it easier to see where macroeconomic and news-based inputs help and where their effects remain limited. In this study, their clearest value

is not perfect point forecasting, but better directional understanding, better turning-point detection, and a more interpretable view of which outside conditions coincide with changes in energy prices.

7 Limitations and Future Work

Energy-price forecasting remains difficult because prices reflect a complicated mix of economic, geopolitical, and market-specific forces, many of which are not fully captured by the selected macroeconomic indicators and aggregated news signals. Although these inputs provide useful context, they may miss sudden shocks or localized events that drive sharp price movements. Classical time-series models are also inherently linear, while neural models, even though they are more flexible, often produce smoother outputs and may still struggle during periods of high volatility.

Another limitation pertains to the representational granularity of the news data. Aggregated GDELT features are computationally convenient, but they necessarily compress event-level detail and may therefore lose some of the richer information embedded in the underlying news stream. In addition, the noise and volatility of energy-price series, especially for natural gas, can affect model stability and make generalization uneven across time periods.

An additional methodological limitation is that we applied walk-forward backtesting only to the LSTM and CNN-LSTM models, whereas we evaluated the classical ARIMA, SARIMA, and SARIMAX specifications on a single fixed holdout split. This creates an asymmetric comparison, because we assessed the neural models under repeated rolling-window re-estimation while we judged the classical models under a one-shot forecasting setup. We did not include the classical models in the backtesting stage primarily to keep the experimental workload tractable across the large set of model-feature combinations and because we treated them as single-shot benchmark specifications after order selection and fitting. Even so, this design choice limits strict comparability across model classes and remains a methodological constraint of the present study.

Future work could address these limitations by expanding both the feature space and the modeling strategy. One natural extension is to test additional macroeconomic variables that may be only indirectly related to energy prices but still carry useful information. Future work could substantially improve the quality and specificity of news-based signals by replacing aggregated GDELT summary statistics with richer natural language representations — such as transformer-derived sentence embeddings (e.g., BERT, FinBERT) — applied directly to the source text of relevant news articles. It may also be useful to model the temporal persistence of events more explicitly, since the effect of a news shock is unlikely to disappear immediately. Finally, attention-based or transformer architectures may be worth exploring, although any gains would still need to be checked carefully against the size and frequency of the available data.

8 Conclusion

This study investigated the extent to which macroeconomic indicators sourced from the FRED database and event-driven news signals derived from the GDELT repository

can meaningfully improve the forecasting accuracy and directional tracking of WTI crude oil and Henry Hub natural gas prices. The results suggest that both sources of information are useful, especially when they are used together in a multivariate framework. Across the experiments, the combined macro-plus-news setting generally performed better than the macro-only and news-only alternatives, which points to a genuinely complementary relationship between structured economic signals and event-driven information.

The findings also suggest that GDELT-based features and macroeconomic variables are more helpful for detecting broad market direction and turning points than for producing consistently precise point forecasts. In particular, the combined-feature SARIMAX models deliver the most reliable overall performance in this study, which suggests that the joint feature space contains explanatory relationships that are harder to recover when the inputs are separated. From an economic perspective, that result is consistent with the idea that monthly oil and gas prices continue to move with persistent macro-financial conditions, while aggregated news signals add timely information about stress and event risk. The neural models capture parts of the underlying dynamics as well, but their forecasting and backtesting behavior is less stable, especially when price changes are abrupt.

Taken together, our empirical results indicate that incorporating heterogeneous exogenous information materially improves directional energy price modeling and reduces forecast error in well-specified linear frameworks, while simultaneously confirming that precise short-term point forecasting of volatile commodity prices remains an open and non-trivial research problem. Energy markets are shaped by rapidly changing geopolitical events, structural breaks, and nonlinear interactions that our aggregated macroeconomic indicators and news summaries represent only partly. For that reason, we view the framework proposed here as a useful step toward richer predictive modeling rather than a complete solution.

This study contributes a structured comparative framework for assessing univariate and multivariate forecasting models under a common validation design, while demonstrating the analytical value of integrating structured macroeconomic indicators with event-driven news features. Within this framework, RMSE serves as an indicator of how closely a model reproduces realized price levels, whereas directional accuracy (DA) reflects how reliably the model captures the sign of period-to-period price changes. Taken together, these evaluation perspectives suggest that external information can strengthen both point-forecast performance and directional inference. At the same time, the results reaffirm that precise forecasting of highly volatile commodity prices remains inherently challenging and continues to warrant further methodological development.

Data Availability and Code Availability

The raw price and macroeconomic series used in this study come from publicly accessible sources, including FRED and GDELT. The codebase and the processed data used for this manuscript is publicly available through the project GitHub repository: <https://github.com/AswinBalajiTR/spring-2026-group6>.

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